

# PipeFit Pro — User Guide

Everything in this app does one job: turn what you can **measure** into what you have to **cut**.

This guide goes module by module. For each one: what it is for, what to measure, what to type, what comes back, **why it comes back that way**, and how to check it against your own head before you cut steel.

## Part 1 — Read this first

### The one thing the whole app is about

A drawing gives you **centre-to-centre**. A saw needs **cut length**. They are never the same number, because a fitting occupies some of the distance between two centres.



**Cut = centre-to-centre – takeout A – takeout B – any weld gaps.**

Every module in this app is that sentence, worked for a different shape of job. Once you see it, the app stops being ten calculators and becomes one.

### Where the numbers come from

Nothing here is a rule of thumb someone remembered.

| Kind of number     | Source                                                           |
|--------------------|------------------------------------------------------------------|
| Fitting takeouts   | The Lindsey Pipefitters Handbook tables, transcribed and checked |
| Pipe OD and wall   | ASME B36.10M                                                     |
| Butt weld fittings | ASME B16.9                                                       |
| Flanges            | ASME B16.5                                                       |
| Screwed fittings   | ASME B16.3                                                       |
| Pipe threads       | ASME B1.20.1                                                     |
| Miter code limit   | ASME B31.3 §304.2.3                                              |
| Bend geometry      | The bend radius, worked exactly                                  |

**1,370 automated tests** run on every change. Ten misprints were found in the printed handbook while transcribing it, and each one is pinned by a test so it cannot quietly come back.

## Why you can trust it without second-guessing

The app is checked against the multipliers a fitter already carries in his head. Multiply the offset by these:

| Fitting angle | Travel         | Run            | Shrink         |
|---------------|----------------|----------------|----------------|
| 11.25°        | × 5.126        | × 5.027        | × 0.098        |
| 22.5°         | × 2.613        | × 2.414        | × 0.199        |
| 30°           | × 2.000        | × 1.732        | × 0.268        |
| 45°           | × <b>1.414</b> | × <b>1.000</b> | × <b>0.414</b> |
| 60°           | × 1.155        | × 0.577        | × 0.577        |
| 90°           | × 1.000        | × 0            | × 1.000        |

The app agrees with every one of them to **four decimal places**, and a test proves it on every build. If it ever disagrees with that table, the app is wrong — not the table.

## Set these once

**Settings** (gear, top right of any screen).

| Setting        | Set it to                                            | Why                                                                                                           |
|----------------|------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| Default size   | The pipe you are on today                            | Every screen opens on it. Saves a tap on every job.                                                           |
| Weld gap       | Your standard root gap, default 3/32"                | Comes off welded and flanged ends only. A screwed or soldered joint pulls up tight and never has it deducted. |
| Fractions      | 1/16 for most work, 1/32 or 1/64 for tight tolerance | The decimal above the fraction is always the exact figure. Cut to the decimal when it matters.                |
| Stock length   | What comes off the rack, default 20'                 | The cut list packs onto it. Get it wrong and the stick count is wrong.                                        |
| Length readout | Inches                                               | The calculator stays in inches. Press <b>FT</b> when you actually want feet.                                  |

# Part 2 — The three ideas everything is built on

# 1. Takeout — why a fitting "eats" length

An elbow does not start at the centre point. The centre point is where two lines cross in the air; the fitting's face is some distance back from it. That distance is the **takeout**.

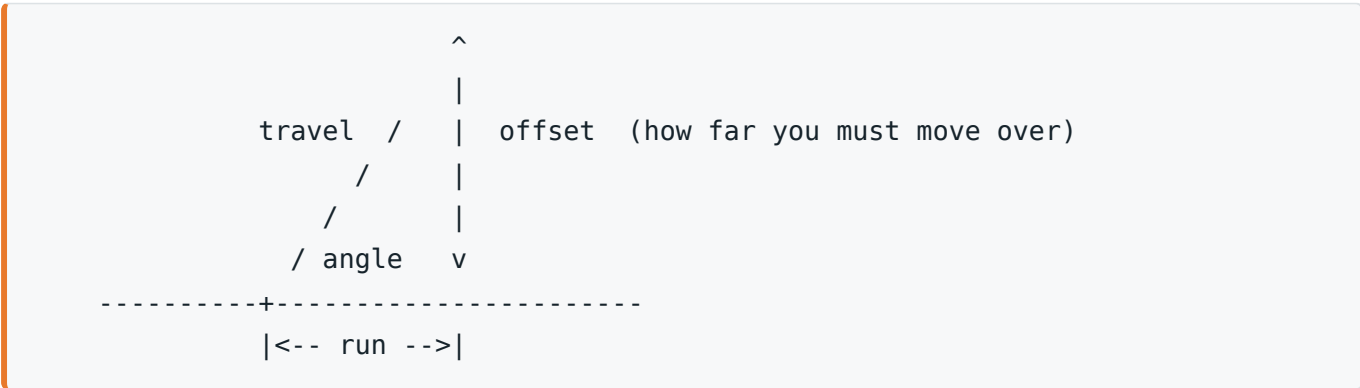
For a **bend you make yourself**, the takeout is pure geometry:

$$\text{takeout} = \text{bend radius} \times \tan(\text{half the angle})$$

A 2" long radius elbow bends on a 3" radius (1.5 × the size). At 90°, tan 45° = 1, so the takeout is **3"**. At 45°, tan 22.5° = 0.4142, so it is **1.2426"**.

For a **fitting you buy**, the takeout is whatever the manufacturer made it, out of the handbook table.

# 2. Every offset is a right triangle



- **Offset** — how far the pipe has to shift. You measure this.
- **Run** — how far it advances along the original line while it shifts.
- **Travel** — centre to centre of the two fittings. **This is the number the cut comes from.**
- **Shrink** — travel minus run. What the offset costs you in length.

Pick any two and the third follows. That is all the offset screens do.

# 3. A bend is NOT a bought fitting

They agree at 90° and disagree everywhere else.

| 2" long radius, 45° | Takes out |
|---------------------|-----------|
| Pipe you bent       | 1.2426"   |
| Elbow you bought    | 1-3/8"    |

A run of bought fittings worked to the bend formula cuts **every piece an eighth of an inch long**.

**The app keeps them separate on purpose. Cut length** offers both, labelled. **Simple offset, Rolling offset** and **Pipe bend** work the bend, because that is what a bender does.

# Part 3 — The modules

## Simple offset

**Get around an obstruction in one plane.** The pipe moves one way — up, down, left or right — and carries on.

### What to measure

- 1. **Offset** — how far the pipe must shift to clear the obstruction, centre to centre.
- 2. Either the **fitting angle** you intend to use, or the **run** you have room for. Not both — the app solves whichever you do not give it.

### Step by step

- 1. Tap **Simple offset**.
- 2. Set the pipe with the wrench button at the bottom (size, LR/SR, schedule).
- 3. Type the **Offset**.
- 4. Tap an angle preset — **45°** is the one you will use most.
- 5. Read **PIPE CUT**.

### What each number means

| Result                   | What it is                           | Why you care                                |
|--------------------------|--------------------------------------|---------------------------------------------|
| PIPE CUT                 | The piece between the two fittings   | This is what you cut                        |
| Travel                   | Centre to centre of the two fittings | The cut before deductions                   |
| Run                      | How far it advances along the line   | Tells you if it fits the space you have     |
| Shrink                   | Travel – run                         | What the offset costs you in overall length |
| Setback                  | Takeout per fitting                  | What comes off each end                     |
| Cut angle                | The angle the fittings turn through  | Confirms the preset took                    |
| Inside arc / Outside arc | Throat and back of the bend          | Marks for bending, not for buying           |

### Worked example

**2" schedule 40, long radius, 12" offset at 45°, default 3/32" gap:**

| Figure             | Value           |
|--------------------|-----------------|
| Run                | 12.0000"        |
| Travel             | 16.9706"        |
| Shrink             | 4.9706"         |
| Setback (each end) | 1.2426"         |
| <b>PIPE CUT</b>    | <b>14.2978"</b> |

**Check it in your head:**  $12 \times 1.414 = 16.97$  travel ✓. Two takeouts of 1.24 = 2.49.  $16.97 - 2.49 = 14.48$ , less two 3/32" gaps (0.19) = **14.29** ✓

### The square jog — 90°

Preset **90°**. The pipe goes straight across and advances nothing. **Run = 0, travel = the offset.**

**24" offset, 2" pipe:** run 0, travel 24.00, setback 3.00 each end → **PIPE CUT 17.81"**

### When the run is fixed instead

Tap **Set run** and type the run. The angle falls out of it and shows as **Cut angle**. A run of **0** gives you the square jog.

Use this when the space along the line is what is tight, not the angle.

### Where people go wrong

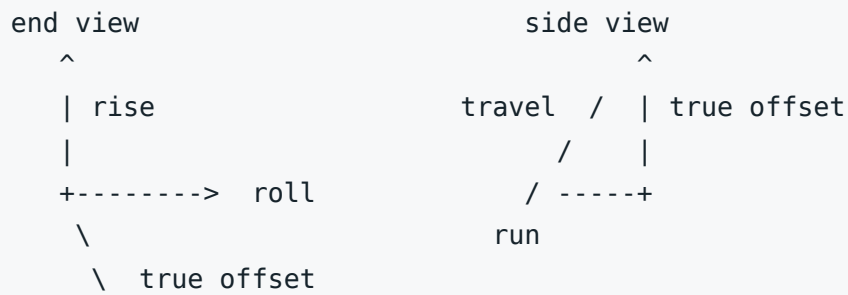
- **Measuring the offset to the pipe surface instead of the centre.** Everything in this app is centre to centre.
- **Forgetting the gap is already off.** PIPE CUT is ready to cut. Do not deduct again.
- **Using a 45° preset while installing bought 45° elbows.** This screen works a *bend*. For bought fittings use **Cut length**, which has both.

## Rolling offset

**The obstruction makes the pipe move two ways at once** — up *and* over. A rolling offset is a simple offset that has been rotated around the pipe.

### Why it is two triangles, not one

Look down the pipe from the end. The **rise** (up/down) and the **roll** (left/right) are two legs of a right triangle. Their hypotenuse is the **true offset** — the real, straight-through distance the pipe has to move.



Then the **true offset** and the **run** form a second right triangle whose hypotenuse is the **travel**. Same theorem twice. That is the whole of it.

## Step by step

1. Tap **Rolling offset**.
2. Set the pipe.
3. Type **Rise / set** (up or down) and **Roll** (left or right).
4. Tap an **Elbow angle** preset, or tap **Run** to fix the run instead.
5. Read **PIPE CUT**.

## What each number means

| Result             | What it is                                                               |
|--------------------|--------------------------------------------------------------------------|
| <b>True offset</b> | The real diagonal distance, straight through                             |
| <b>Roll angle</b>  | How far around the pipe the offset lies — which way to point the fitting |
| <b>Travel C2C</b>  | Centre to centre of the two fittings                                     |
| <b>Run</b>         | Advance along the original line                                          |
| <b>PIPE CUT</b>    | The piece between them                                                   |

**The roll angle is the one people miss.** It is not a cut dimension — it tells you how far to rotate the fitting around the pipe so the offset lands where you measured it.

## Worked example

**2" schedule 40 LR, rise 12", roll 9", 45° elbows:**

| Figure          | Value           |
|-----------------|-----------------|
| True offset     | 15.0000"        |
| Roll angle      | 36.87°          |
| Run             | 15.0000"        |
| Travel C2C      | 21.2132"        |
| Setback         | 1.2426"         |
| <b>PIPE CUT</b> | <b>18.5404"</b> |

**Check it:** 12 and 9 is a 3-4-5 triangle scaled by three — true offset **15** ✓. At 45° the run equals the true offset, 15 ✓. Travel =  $15 \times 1.414 = \mathbf{21.21}$  ✓

## Where people go wrong

- **Entering the true offset in the rise box.** Rise and roll are the two *separate* measurements. The app works the true offset out.
  - **Ignoring the roll angle.** Get it wrong and the spool is geometrically perfect and points the wrong way.
  - **Rise and roll both zero** — there is no offset to solve, and the app says so.
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## Cut length

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**The most used screen in the app.** You have a centre-to-centre dimension off a drawing and a fitting on each end.

### Step by step

1. Tap **Cut length**.
2. Set the pipe.
3. Type **C2C length**.
4. Pick **Joint** — screwed, welded, flanged or soldered.
5. If flanged, pick the **Class** — 150, 300, 600 and up.
6. Pick **End A** and **End B**.
7. Read **PIPE CUT**.

### Why the joint family matters

It changes two things at once:

1. **Which table the takeout comes from.** A screwed 90 and a welded 90 in the same size are different castings with different dimensions.
2. **Whether the weld gap is deducted.** Welded and flanged ends leave a root gap. Screwed and soldered joints pull up tight and never have one deducted.

That second one was a real bug once. It is now a rule in the code and a test.

### Worked example

**24" C2C, 2" schedule 40, welded long radius 90 on both ends:**

| Figure          | Value           |
|-----------------|-----------------|
| End A takeout   | 3.0000"         |
| End B takeout   | 3.0000"         |
| Total deduction | 6.0000"         |
| <b>PIPE CUT</b> | <b>18.0000"</b> |
| Weight          | 5.48 lb         |

With a 1/8" gap on both ends: **17.75"**

### Anything not on the list

Pick **Custom** on that end and type the takeout yourself. Use it for a valve, a strainer, a specialty fitting, or anything a vendor drawing gives you directly.

### Where people go wrong

- **Leaving the joint on the wrong family.** Check the **Joint** chip before you read the number.
- **Assuming a 45 is a 45.** A bought 45 and a bent 45 have different takeouts. Both are on the list, labelled.
- **Not setting the class on flanged.** A 150 and a 300 flange are not the same length.

## Thread engagement

**How far a threaded joint pulls up**, and what that does to your cut.

### Why a threaded joint is different

On a welded joint the pipe stops at the fitting face. On a threaded joint the pipe **screws into** the fitting — so the pipe reaches past the face, and the deduction is smaller than the fitting's centre-to-face.

**Deduction per end = fitting centre-to-face – total engagement**

Engagement has two parts:

- **Hand tight (L1)** — how far it goes in by hand. From ASME B1.20.1.
- **Wrench makeup** — how far your wrench turns take it further. Each turn advances the pipe by one **pitch**, and pitch is  $1 \div \text{threads per inch}$ .

### Step by step

1. Tap **Thread engagement**.
2. Tap the **Size**.
3. Set **Wrench turns** if you pull up tighter or looser than standard.
4. Type **C2C length** to get the cut.

## Worked example

1" NPT, standard 3 wrench turns, 24" centre to centre:

| Figure                   | Value           | Where it comes from    |
|--------------------------|-----------------|------------------------|
| Threads per inch         | 11.5            | from the standard      |
| Pitch                    | 0.0870"         | $1 \div 11.5$          |
| Hand tight (L1)          | 0.4000"         | from the standard      |
| Wrench makeup            | 0.2609"         | $3 \times 0.0870$      |
| <b>Total engagement</b>  | <b>0.6609"</b>  | hand tight + wrench    |
| Thread remaining         | 0.3236"         | what is left unused    |
| Fitting centre-to-face   | 1.5000"         | B16.3 90° elbow        |
| <b>Deduction per end</b> | <b>0.8391"</b>  | $1.5000 - 0.6609$      |
| <b>PIPE CUT</b>          | <b>22.3217"</b> | $24 - 2 \times 0.8391$ |

**Watch "Thread remaining".** If it goes negative the joint is over-threaded — the pipe bottoms out before it seals.

## Where people go wrong

- **Using the welded takeout on screwed pipe.** It is a different, bigger number and the piece comes out short.
- **Not adjusting wrench turns to how you actually pull up.** Two extra turns on 1" pipe moves the cut by about 3/16" over two ends.

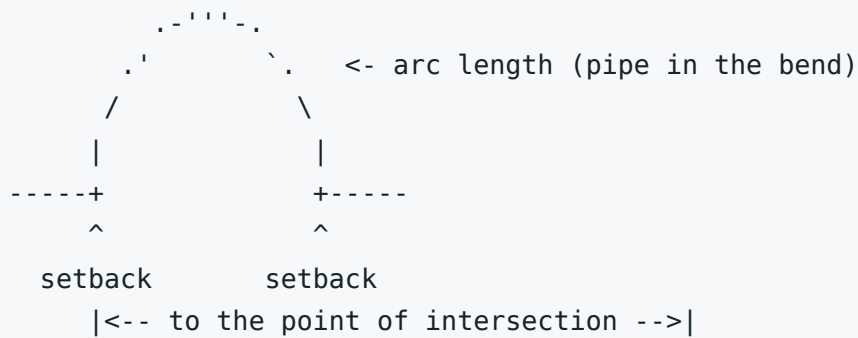
## Pipe bend

**Any angle, any radius, on your own bender.** This is where the marks come from.

### Why bending needs different numbers from cutting

When you bend, the pipe does not go round the corner — it cuts the corner on an arc. Three numbers describe that:

- **Setback** — from the point where the two legs would cross, back to where the bend starts.  $\text{radius} \times \tan(\text{half the angle})$ .
- **Arc length** — how much pipe is actually inside the bend.
- **Gain** — tangent total minus arc. **What you save by bending instead of mitring the corner.** Forget it and every piece comes out long.



## Step by step

1. Tap **Pipe bend**.
2. Pick the **Size**, then a **Radius** rule — 3D tight, 5D common field minimum, 10D long sweep — or type a radius.
3. Type the **Angle**.
4. Type **Leg A** and **Leg B**, measured **to the point of intersection**.
5. Set **Stock** if you want to know how many legs come out of a length.
6. Read the marks.

## Worked example

**2" pipe, 5D radius (10"), 45°, legs 30" and 30":**

| Figure                | Value           |
|-----------------------|-----------------|
| Setback (per tangent) | 4.1421"         |
| Arc length            | 7.8540"         |
| Gain                  | 0.4303"         |
| Tangent total         | 8.2843"         |
| <b>Mark from end</b>  | <b>25.8579"</b> |
| End of bend           | 33.7118"        |
| Piece length          | 59.5697"        |

**Check it:** setback =  $10 \times \tan 22.5^\circ = 4.142$  ✓. Mark =  $30 - 4.142 = 25.858$  ✓. Arc =  $10 \times 45^\circ$  in radians =  $10 \times 0.7854 = 7.854$  ✓

**At 90°, 5D radius, legs 24" and 18":** setback 10.00, arc 15.71, gain 4.29, mark from end 14.00, piece 37.71.

## Springback

If your bender overbends, put your known springback in and the app gives you the **overbend angle** to pull to. Leave it at zero if you have already dialled it out.

# Saddle bend

**Jump over a pipe or a beam without changing the line the conduit runs on.** It goes up, over, and back down to exactly where it was.

## Three point or four point

- **Three point** — for a round obstruction. One centre bend with two half-angle bends either side.
- **Four point** — for a wide obstruction. Two offsets back to back with a flat run over the top. Needs the **Width**.

## Why the centre bend is double the side bends

The conduit has to come back to the same line it started on. If the centre bend turns it through 45°, each side bend must turn it back through half that — 22.5° — for the two ends to end up parallel.

## Step by step

1. Tap **Saddle bend**.
2. Pick **Three point** or **Four point**.
3. Type the **Depth** — how far it has to rise to clear.
4. Type **To obstruction** — from the conduit end to the centre of the obstruction.
5. On a four point, type the **Width**.
6. Pick the centre bend angle.
7. Bend at the marks, in order.

## Worked example

**Three point, 2" deep, 20" from the end, 45° centre bend:**

| Figure               |  | Value    |  |
|----------------------|--|----------|--|
| Side bend angle      |  | 22.5°    |  |
| Multiplier           |  | 2.6131   |  |
| Shrink per bend pair |  | 0.3978"  |  |
| Total shrink         |  | 0.7956"  |  |
| Developed length     |  | 10.4525" |  |
| Minimum clearance    |  | 4.8284"  |  |

| Mark   | At      | Bend  | Note                              |
|--------|---------|-------|-----------------------------------|
| Mark 1 | 15.172" | 22.5° | Bend up, arrow to the centre mark |
| Centre | 20.398" | 45°   | Bend over the obstruction         |
| Mark 3 | 25.624" | 22.5° | Bend down, back to level          |

**Check it:** spacing = depth × multiplier =  $2 \times 2.6131 = 5.226$ ". Centre mark is at  $20 + 0.398 \text{ shrink} = 20.398$ , and the outer marks sit 5.226 either side ✓

**Minimum clearance** is the one to watch. If the obstruction is closer to the end than that figure, the first bend would fall off the end of the conduit — and the app refuses rather than giving you a mark you cannot make.

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## Miter bend

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**Make an elbow out of straight pipe** when you do not have a fitting, or the size does not come as one.

### How the cut angle is worked out

With **n** segments you make **n – 1** cuts. Each cut face turns the pipe by **twice** the angle you cut it at, because both halves tilt. So:

$$\text{cut angle} = \text{total turn} \div (2 \times \text{number of cuts})$$

A 90° elbow in 4 segments is 3 cuts, so each cut is  $90 \div 6 = 15^\circ$ .

The two **end segments** have one mitred face. The **middle segments** have two, so their total included angle is double.

**Throat** is the short side of a segment, **back** is the long side. They come from the centreline radius minus and plus half the pipe OD.

### Step by step

1. Tap **Miter bend**.
2. Type the **Total turn** and pick the **Segments**.
3. Set the **Centreline radius**.
4. Mark and cut throat and back lengths on each segment.

### Worked example

**90° total, 4 segments, 2" schedule 40, 6" centreline radius:**

| Figure           | Value                     |
|------------------|---------------------------|
| Cuts             | 3                         |
| <b>Cut angle</b> | <b>15.00°</b>             |
| End segment      | 15.00° (one mitred face)  |
| Mid segment      | 30.00° (two mitred faces) |
| Throat length    | 2.5790"                   |
| Back length      | 3.8518"                   |
| Cutback          | 0.6364"                   |
| Centreline arc   | 9.4248"                   |

**Check it:** 2" pipe OD is 2.375", so half is 1.1875. Throat =  $2 \times (6 - 1.1875) \times \tan 15^\circ = 2 \times 4.8125 \times 0.2679 = 2.579$  ✓

### The code check

Above a **22.5° cut angle** the app warns you: ASME B31.3 §304.2.3 requires design verification for miters above that. It does not stop you — it tells you, because that is an engineering decision, not a field one.

More segments = smaller cut angle = a smoother, stronger elbow. If you get the warning, add a segment.

## 3D spool

**Build a whole run and look at it before you cut any of it.**

### How a spool is described

A leg is **where it runs and how far**. Nothing else.

- **Length** — centre to centre, the way a spool is dimensioned.
- **Runs** — a point off the compass: N, NE, E, SE, S, SW, W, NW.
- **Rise** — level, straight up, straight down, or a slope in between.

So the spool everybody draws first is *36" east, 24" straight up, 30" north*, and that is exactly what you type.

**You never enter a bend angle and you never enter a roll.** The app works out the turn between one leg and the next, tells you what it is, and tells you whether it comes off a shelf:

Turns 90° off leg 1 — 90° elbow

If a turn is not a stock angle you get told that too, with the nearest stock angle beside it, and a warning above the leg list so it cannot be missed before you order.

### Step by step

1. Tap **3D spool**.

- 2. Set the pipe and the gap.
- 3. Set the first leg's **Length** and where it **Runs**.
- 4. **Add leg**, then give it a length and a direction.
- 5. Read the turn the app worked out under each leg.
- 6. **Hold a leg in the picture and drag along it** to stretch or shorten it.
- 7. Read **TOTAL PIPE** and the per-leg cuts.

The drawing

Every leg on the picture carries **its own length and where it runs**, and every fitting carries **its angle**. The picture is the drawing — you do not have to look anything up in a list to read it.

Nine views, under the drawing:

| View               | What it is                                                   |
|--------------------|--------------------------------------------------------------|
| NE NW SW SE        | The four isometric corners.                                  |
| Plan               | Looking straight down, north up the page, east to the right. |
| S / E / N / W elev | Square-on elevations from each side.                         |

It opens on the corner this spool reads best from. Of the four isometric corners, the app picks the one where no two legs cross on the page and nothing is lost end-on — the same spool can read as a clean L from one corner and as a closed triangle from another, and the second one is a puzzle, not a drawing. Change the shape and it picks again; turn it by hand or pick a plan and it leaves your view alone.

Drag anywhere to turn it. Pull right and it goes right, pull down and the top comes over. Every angle is reachable, right over to looking straight down.

A leg drawn as a circle is a leg coming at you. In a plan a riser has no length on the page, and in an elevation neither does a leg running toward you. That is not the drawing failing — it is what a plan is, and it is why the figure beside it says 24" UP . Read the figure.

Turning the spool over

Three controls, and **not one of them changes a cut**. All three are reflections, and a reflection keeps every angle it finds — so the bends are the same bends, the takeouts are the same takeouts, and the pipe you buy is the same pipe.

| Control   | What it does                                                                                       |
|-----------|----------------------------------------------------------------------------------------------------|
| Mirror    | The opposite hand. The first leg stays where it was; everything after it comes off the other side. |
| Turn over | Every rise becomes a drop. What ran up now runs down.                                              |
| Swing 90° | The whole spool a quarter turn round the compass. Same shape, facing elsewhere.                    |

**Mirror** is the one you want when the same spool is needed handed for the other side of a rack. **Turn over** is the one you want when the whole run should fold the other way. **Swing** is for when the shape is right and it is simply pointing the wrong way on the drawing.

## Worked example

2" schedule 40 LR, 36" east then 24" straight up then 30" north, 3/32" gap:

| Figure           | Value       |
|------------------|-------------|
| Centre to centre | 90.00"      |
| Total pipe       | 77.63"      |
| Elbows           | 2, both 90° |

| Leg | C2C    | Cut    |
|-----|--------|--------|
| 1   | 36.00" | 32.91" |
| 2   | 24.00" | 17.81" |
| 3   | 30.00" | 26.91" |

Leg 1 loses one takeout (it has a fitting on one end only). Leg 2 sits between two elbows and loses two. That is the whole difference, and you can see it in the numbers.

### The cut list — what to pull off the rack

Knowing every cut is not the same as knowing what to go and get. Four pieces totalling 180" are not fifteen feet of pipe; they are **one** twenty-foot stick if they nest and **two** if they do not.

Under the elbows the app shows the sticks themselves, drawn to scale, with the leg numbers on the pieces and the drop left blank on the end. Under them, in words: how many to pull, the longest drop, and what the saw took.

Two things it is careful about, and both of them cost pipe:

- **The blade takes pipe.** Every piece severed off a stick costs a **saw cut** as well as its own length, so a stick that looks like it holds exactly four 60" pieces holds three. Set your blade in **Settings** → **Saw cut**. The kerf is charged on the last piece too, which over-states the loss by one cut on a stick used right to the end — the only direction it is safe to be wrong in when you are ordering.
- **It finds the fewest sticks, not a good guess.** The usual way to pack a cut list is biggest first into the first stick that takes it, and it can call for a stick nobody needs — 7, 5, 4, 2, 2 on an 11 ft stick comes out as three that way and fits on two. For a spool the app tries every grouping and takes the best, and among the ones that use the same number of sticks it takes the one that **leaves the longest single drop**, because one long drop is material and the same footage in four short ones is scrap.

If any single cut is longer than a stick, it says so by name rather than quietly planning something that cannot be built.

**The same check on the single-cut screens.** Cut length, simple offset and rolling offset all show the stock length, and now all three say whether the cut comes off one stick and what is left. That figure used to sit there meaning nothing.

## Why the drawing is drawn the way it is

It is a **true isometric** projection — yaw  $-45^\circ$ , pitch  $35.264^\circ$ . That is the only orientation where all three axes are foreshortened equally and sit  $120^\circ$  apart, and it is what makes iso paper read as solid instead of flat.

The drawing is scaled to what is actually on the page, so it **fills the canvas in every view** rather than leaving a quarter of it empty. While your thumb is down the scale is held so the picture cannot swell under you; it refits the moment you let go.

## Calculator

A trade calculator that understands feet, inches and fractions — and **stays in inches**.

### How to enter a dimension

Type the number, **then** the unit. Not the other way round.

| You want      | Press               |
|---------------|---------------------|
| 30 inches     | 3 0 Inch            |
| 6 feet        | 6 Feet              |
| 30-1/2 inches | 3 0 Inch 1 / 2 Inch |
| 6 foot 3      | 6 Feet 3 Inch       |

### It stays in inches

30 Inch  $\times$  6 = gives **180"**, not 15 feet. The **IN** marker top left tells you which unit you are reading.

- **Feet** converts what is on display to feet and inches — 2' 6". The marker changes to **FT-IN**.
- **Inch** takes it back.
- The choice sticks until you change it, and survives a **Clear**.

### The trade keys

The dark keys along the top are the pipefitting ones. **Conv** turns on the second function printed above each key.

| Key                        | What it does                                |
|----------------------------|---------------------------------------------|
| Offset, Run, Travel        | Enter any two, the third solves             |
| Angle/Slope                | Angle or slope of a run                     |
| Pipe Size, Pipe Mat'l      | Set the pipe for the trade keys             |
| Circle, $x^2$ , $\sqrt{x}$ | Area and diameter work                      |
| Store / Rcl                | Hold a figure while you work something else |

### Where people go wrong

- **Pressing the unit first**. It does nothing. Number, then unit.

- **Reading the fraction instead of the decimal.** The fraction is rounded to your setting. The decimal is exact.

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## Flange bolt-up

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Work the bolts in the order the standard says, and have the app refuse the wrong one.

### Why a flange is not tightened round the circle

A flange joint is a spring. Pull one bolt down hard and the flange tips towards it, which *unloads* the bolts on the far side. Go round the circle in order and you chase that tip all the way round: by the time you get back to bolt 1 it is slack, the gasket is crushed on one side and dry on the other, and the joint weeps.

Two rules fix it, and this module is nothing more than those two rules made hard to get wrong:

- **Cross the flange.** Every bolt is followed by the one **straight across** from it, so the two sides come down together and the flange stays parallel.
- **Do it in passes.** Snug the whole joint at about a third, go round again at about two thirds, then again at full. Tightening any bolt relaxes its neighbours, so the first trip round is always uneven no matter how careful you are.

Then one more that gets skipped more than any other step in pipefitting:

- **The last pass goes round, not across, at full torque.** Its job is to pick up the relaxation the three cross passes left behind. This is the *rotational* pass in ASME PCC-1, and it is the difference between a joint that holds and a joint that needs re-torquing next week.

### The four passes

| Pass | Torque | Order        | What it is doing                                                 |
|------|--------|--------------|------------------------------------------------------------------|
| 1    | ~30%   | Across       | Snugging the joint up square before anything is pulled down hard |
| 2    | ~60%   | Across       | The gasket is seating, so bolts done early have gone slack       |
| 3    | 100%   | Across       | Full torque, still crossing                                      |
| 4    | 100%   | <b>Round</b> | The check. Picks up what the cross passes left                   |

Four passes over every bolt. On an 8 bolt flange that is 32 taps; on a 16 bolt flange, 64.

### Step by step

1. Tap **Flange bolt-up**. It opens on the size the app is set to.
2. Set **CLASS** (125 or 250 lb) and **SIZE**. The bolt count follows the size — you do not set it.
3. If the flange is not in the cast iron tables (a raised-face steel flange, say), set **BOLTS** directly. The sequence only needs the count.
4. Optional: put the **final torque from the job's bolting spec** in the torque field. The header then shows the figure for each pass instead of the percentage.
5. The header says which bolt is next. Find it on the face — it has a dark ring around it — and tighten it.

- 6. **Tap that bolt.** It changes colour and the header moves to the next one.
- 7. Work round until the header reads **Joint complete**.

What happens if you tap the wrong bolt

Nothing moves. The app tells you which bolt you hit, which bolt it wants, and flashes red on the **right** one. Nothing is recorded, no colour changes, and the sequence does not advance.

That is the point. A bolt map you can tap anywhere is a picture. One that only accepts the bolt the sequence is asking for is a check.

**Undo** takes the last bolt back, including back into the previous pass if that is where you are. **Start over** clears the joint.

What the colours mean

| Colour       | Meaning                    |
|--------------|----------------------------|
| Grey outline | Not touched yet            |
| Yellow       | Snug, pass 1 at ~30%       |
| Orange       | Pass 2 at ~60%             |
| Blue         | Pass 3, full torque        |
| Green        | Checked on the final round |

Four flat, saturated colours rather than the soft tints the rest of the app uses, because you have to tell a bolt at a third from a bolt at two thirds *across the width of a phone, in daylight*.

Worked example — 6" class 125

Eight bolts. The app shows this from the table:

| Figure           | Value                 |
|------------------|-----------------------|
| Flange OD        | 11"                   |
| Flange thickness | 1"                    |
| Bolt circle      | 9-1/2"                |
| Bolts            | 8 × 3/4", 3-1/4" long |
| Bolt hole        | 7/8" drilled          |
| Bolt to bolt     | 3.6355"               |
| Ring gasket      | 6" × 8-3/4"           |

Passes 1, 2 and 3 (across):

1 → 5 → 3 → 7 → 2 → 6 → 4 → 8

Pass 4 (round):

1 → 2 → 3 → 4 → 5 → 6 → 7 → 8

With a bolting spec of **300 ft-lb**, the app shows **90**, then **180**, then **300**, then **300 ft-lb**.

Read the cross sequence and you can see the rule: 1 then 5 is straight across (180°), 5 then 3 is a quarter turn back, 3 then 7 straight across again. Every *pair* is opposite; the pairs themselves step round the flange.

### Worked example — 16" OD class 125

Sixteen bolts, and the sequence a lot of shops have taped inside a gang box:

1 → 9 → 5 → 13 → 3 → 11 → 7 → 15 → 2 → 10 → 6 → 14 → 4 → 12 → 8 → 16

The app generates this rather than looking it up, by the halving rule the tabulated sequences are built on: take the pattern for half the bolts and follow each entry with the bolt directly opposite. It reproduces the published **4, 8, 12 and 16** bolt orders exactly, which is the check that the rule is the right one.

### Where the bolt numbers are on the real flange

Bolt 1 is the first hole **clockwise of top dead centre**, and it sits **half a pitch off the centreline, not on it**. That is how flanges are drilled — it is what lets a fitting be turned a quarter turn and still bolt up — and it is how the app draws them.

- 8 bolts: first hole at 22.5°, then every 45°
- 16 bolts: first hole at 11.25°, then every 22.5°
- 24 bolts: first hole at 7.5°, then every 15°

Any starting bolt works as long as you keep the pattern. The app starts at its bolt 1; if you would rather start somewhere else, turn the screen so the app's bolt 1 lands on yours and keep going round the same way.

### Big flanges

A 48" flange has 44 bolts and a 96" has 68. Drawn to fit a phone those holes would be twelve pixels apart, and a touch target that spills into its neighbour would register the wrong bolt — which on a screen whose whole job is to refuse the wrong bolt is worse than useless.

So above about 40 bolts the face **grows and scrolls sideways** instead of the bolts shrinking into each other. Swipe the picture across. Every target stays at least 26 pixels from the next one, at every count in both tables.

### Where people go wrong

- **Skipping the last pass.** Three cross passes and no check round is the most common version of this job, and it is the one that comes back.
- **One bolt straight to full torque.** Tempting when you are near the end of a shift. It cocks the flange and unloads everything opposite.
- **Guessing the torque.** The app deliberately does not supply a final figure. It depends on the gasket, the stud material and whether the threads are lubricated, and a guess either crushes the gasket or leaves the joint loose. Get it from the bolting spec.

- **Treating "Joint complete" as a sign-off.** The app recorded a sequence. It did not measure torque and it cannot see the gasket. On anything that runs hot, check the joint again after the line has been up to temperature.

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## Joint register

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Every bolt-up is saved as you work it, bolt by bolt.

### What it is for

A flange bolt-up is four passes over every bolt. On a real job you get called away in the middle of one — the crane needs a hand, the welder wants the fit checked, your phone goes in your pocket and stays there.

Come back and you have two bad options: start the joint again, or guess which bolts you had already pulled down. **Guessing is how a bolt gets taken to full torque twice while its neighbour never gets touched at all** — which is exactly the uneven joint the passes exist to prevent.

So the app writes the joint down **on every bolt**, not when you leave the screen. Close the app, kill it, restart the phone: you come back on the same bolt, with the same bolts coloured in.

### Two kinds of joint

|                   | Where it lives                            | When to use it                                         |
|-------------------|-------------------------------------------|--------------------------------------------------------|
| The unnamed joint | Opens straight from <b>Flange bolt-up</b> | One joint, no paperwork. Still saved, still resumable. |
| A named joint     | Kept in the register by its tag           | You will come back to it, or you want a record of it   |

You never have to name anything. Open **Flange bolt-up**, work the joint, and it is saved either way. Naming is for when you want it **kept** — several joints on the go, or a record that the joint was done.

### Step by step

1. Tap **Flange bolt-up** and work the joint as normal. The bar at the top reads **Unnamed joint · Saved as you go**.
2. To keep it, tap **Name it**. Give it whatever you would call it out by — a line number, a spool mark, a valve tag — and a note for where it is.
3. It moves into the register under that tag, **with the work you have already done**, and the unnamed slot is cleared for the next one.
4. Tap the **list icon** in that bar, or **Joint register** on the home screen, to see everything.
5. Tap any joint to pick it up exactly where it was.

## Reading a row

8-CWS-102 FL-3  
6" · class 125 · 8 bolts  
— — — — Pass 2 · bolt 5 of 8  
north rack, behind the pump  
Worked 20 min ago

The four bars are the four passes, filled as far as the joint has got, in the **same colours the flange face uses** — yellow, orange, blue, green. One glance tells you how far round it is without opening it.

## How the list is ordered

- **Working now** — the unnamed joint, if there is anything in it.
- **Part done** — live joints, most recently worked at the top. This is where you look when you come back from break.
- **Finished** — all four passes recorded, most recently finished first.

Delete asks first, in the row. On a joint that is only part done it says so before it goes.

## Changing the flange starts the bolt-up again

Pick a different size, class or bolt count on a joint and **the bolt-up resets**. That is deliberate: a level array for sixteen bolts means nothing on a flange with twelve, and carrying it across would put colours on bolts nobody touched.

Setting the **BOLTS** count by hand unsets the size, because a hand-set count belongs to no table row. The picture falls back to generic proportions rather than drawing a flange it is not.

## What it will not do

Two things it refuses on purpose, because both come down to the same rule — **the app would rather lose a joint than show you a wrong one**.

**A stored joint that does not add up is not loaded.** Every saved bolt-up is checked against itself on the way in: in any state the app can actually reach, every bolt sits at either the current pass or one above it, and the number of bolts one above is exactly how far through the pass you are. If a stored joint fails that, it is dropped and the register says how many went. A half-repaired bolt-up state looks exactly like a real one on screen, and would put you on the wrong bolt.

**A list written by a newer version of the app is left alone.** If the register was last written by a newer build — the web app and the APK do get out of step — nothing is saved at all, and the screen says so. The alternative is the older build quietly overwriting joints the newer one is holding. If you would rather start clean, **Start new** on that notice does it, and says what it costs.

## Where people go wrong

- **Assuming you have to name it.** You do not. The unnamed joint is saved the same as any other.
- **Naming every joint.** The register is for joints you will come back to. One-and-done joints do not need a tag.

- **Expecting it to sync.** It does not. The register lives on **that phone**. Two people on the same flange keep two lists.
- **Treating "Finished" as a sign-off.** It records that the sequence was followed. It did not measure torque and cannot see the gasket.

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## Re-torque log

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A joint that was right cold can be slack hot. This records what you find when you go back.

### Why a joint needs checking again

A bolted joint is a spring holding a gasket squashed. Take the line up to temperature and three things happen at once:

- The **gasket creeps** under load and heat, and gives up some of its thickness for good.
- The **bolts and flanges grow at different rates**, because they are different steels in different sections.
- The whole joint **relaxes**, and what was 300 ft-lb cold is less than that hot.

None of it is a mistake anybody made. It is what the joint does. Which is why most bolting specs on hot service say to go round it again once it has been through a thermal cycle — and why that pass is the one that actually keeps the joint.

### The one thing worth writing down

Not the date. **Whether anything moved.**

A date on its own says somebody went back. It does not say what they found, and what they found is the whole reading:

| What you found       | What it means                                              |
|----------------------|------------------------------------------------------------|
| <b>Bolts took up</b> | Still relaxing. Go back to it after another cycle.         |
| <b>All tight</b>     | It has settled. That is the check that closes a joint out. |

So the app asks that question with **two buttons and no default answer**. A preselected answer is one somebody taps past without walking to the flange.

### Step by step

1. Finish the bolt-up. The **Re-torque** section appears under the controls — only on a joint that has been finished once, because there is nothing to re-check otherwise.
2. Once the line has been up to temperature and back, go round the flange **at full torque, in order**.
3. Tap **Record a check**.
4. Answer **Bolts moved** or **All tight**.
5. Optionally add the torque you used and a note — "after 8 hr at temp", "two bolts took a quarter turn".
6. **Save**.

The heading above the log then reads one of three things: *Not checked since it came up to temperature*, *Still taking up*, or *Nothing moved last time*.

## The register sorts itself by what is left to do

| Section          | What is in it                                                     |
|------------------|-------------------------------------------------------------------|
| Working now      | The unnamed joint                                                 |
| Part done        | Bolt-ups still going                                              |
| Needs a re-check | Finished, but never been back to — or still taking up when it was |
| Closed out       | Finished, checked, nothing moved                                  |

**Needs a re-check is the list to work from** after a startup. Finished is not the same as closed out, and the register stopped treating it that way.

**Clear closed out** only ever removes the settled ones. A joint still waiting on a re-check is not finished with, whatever its four passes say.

## What the app will not do

**It will not tell you when to go back.** That depends on the service, the medium, the gasket and the spec — a hard cycle wants checking sooner than a line that warms up once and sits there. Inventing a schedule would be inventing engineering. The app records what you found and shows you what is outstanding; the timing comes from the job.

**A check cannot be recorded on an unfinished joint.** There is nothing to re-check on a bolt-up nobody has been through once.

**Reopening a bolt-up throws the log away.** Undo past the end, or change the size, class or bolt count, and the checks go with it — they were checks of a joint that no longer exists, and keeping them would be a record of something nobody finished.

## Where people go wrong

- **Treating the fourth pass as the end.** It closes the *cold* bolt-up. On hot service the joint has not finished moving yet.
- **Logging the date and not the finding.** "Checked 14 Sep" tells the next person nothing. "Checked 14 Sep, two bolts took a quarter turn" tells them to come back.
- **Closing a joint out on one check where bolts moved.** That check said the opposite — it is still going. The app will keep it under **Needs a re-check** for exactly that reason.
- **Expecting it to sync.** It does not. The log lives on that phone, like the rest of the register.

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## Handbook

Every table from the printed book, searchable, **with its page number on it.**

Use it when you want to see the figure the calculation used rather than take it on trust. Search by what you are looking for — "reducer", "flange", "copper", "support" — and the table comes up with its source page so you can check it against the book in your truck.

## Settings

| Setting        | What it does                                                                                            |
|----------------|---------------------------------------------------------------------------------------------------------|
| Theme          | Light, dark or follow the phone                                                                         |
| Units          | Imperial or metric                                                                                      |
| Fractions      | Off, 1/8, 1/16, 1/32, 1/64                                                                              |
| Length readout | Inches, or feet + inches, for the calculator                                                            |
| Default size   | Pipe every screen opens on                                                                              |
| Weld gap       | Your standard root gap                                                                                  |
| Stock length   | The length you buy. The cut list packs onto it, and the offset and cut-length screens check against it. |
| Saw cut        | What the blade takes, default 1/8". Charged on every cut in the cut list.                               |
| Updates        | What is running, and a manual check                                                                     |

Settings are stored on the phone only. Nothing leaves it.

# Part 4 — Knowing when to trust it

## What is exact and what is not

| Figure                        | Value                                                                                                          |
|-------------------------------|----------------------------------------------------------------------------------------------------------------|
| Exact                         | Every length, angle, takeout, arc and cut. These are geometry and published tables.                            |
| Estimate, and labelled as one | Fitting weights. They are geometric approximations, not vendor catalogue figures. Pipe weight itself is exact. |

## Fractions versus decimals

The fraction is **rounded to the nearest tick** of the denominator you picked. The decimal above it is the exact figure.

On a 10-foot run at 1/16" rounding you can be out by up to 1/32". Usually nothing. On a tight flange-to-flange fit-up it is not nothing. **Cut to the decimal when the tolerance is tight.**

# When the app refuses to answer

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It refuses rather than guessing, and it tells you why:

| Message                                               | What it means                                                                                          |
|-------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| "Fitting takeoffs exceed travel"                      | The fittings are longer than the space. No pipe between them. Use a smaller angle or a shorter radius. |
| "Obstruction is too close to the conduit end"         | The first bend would fall off the end. Move the mark or use a steeper angle.                           |
| "Fitting angle past 90° turns the run back on itself" | Past 90° the pipe doubles back. That is not an offset.                                                 |
| "Cut angle exceeds 22.5°"                             | Code check, ASME B31.3. Not a refusal — add a segment or get it verified.                              |
| "Thread remaining" going negative                     | Over-threaded. The pipe bottoms out before it seals.                                                   |

**A refusal is information.** It is the app telling you the job as measured cannot be built that way — which is much better news before you cut than after.

# The habit worth keeping

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For anything that matters, do this once:

- 1. Read **PIPE CUT**.
- 2. Check **Travel** against the multiplier table in Part 1.
- 3. Check the **takeout** looks right for the fitting in your hand.

Three seconds, and you will never wonder whether the number is good.

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*PipeFit Pro. Built for the job, tested against the book.*